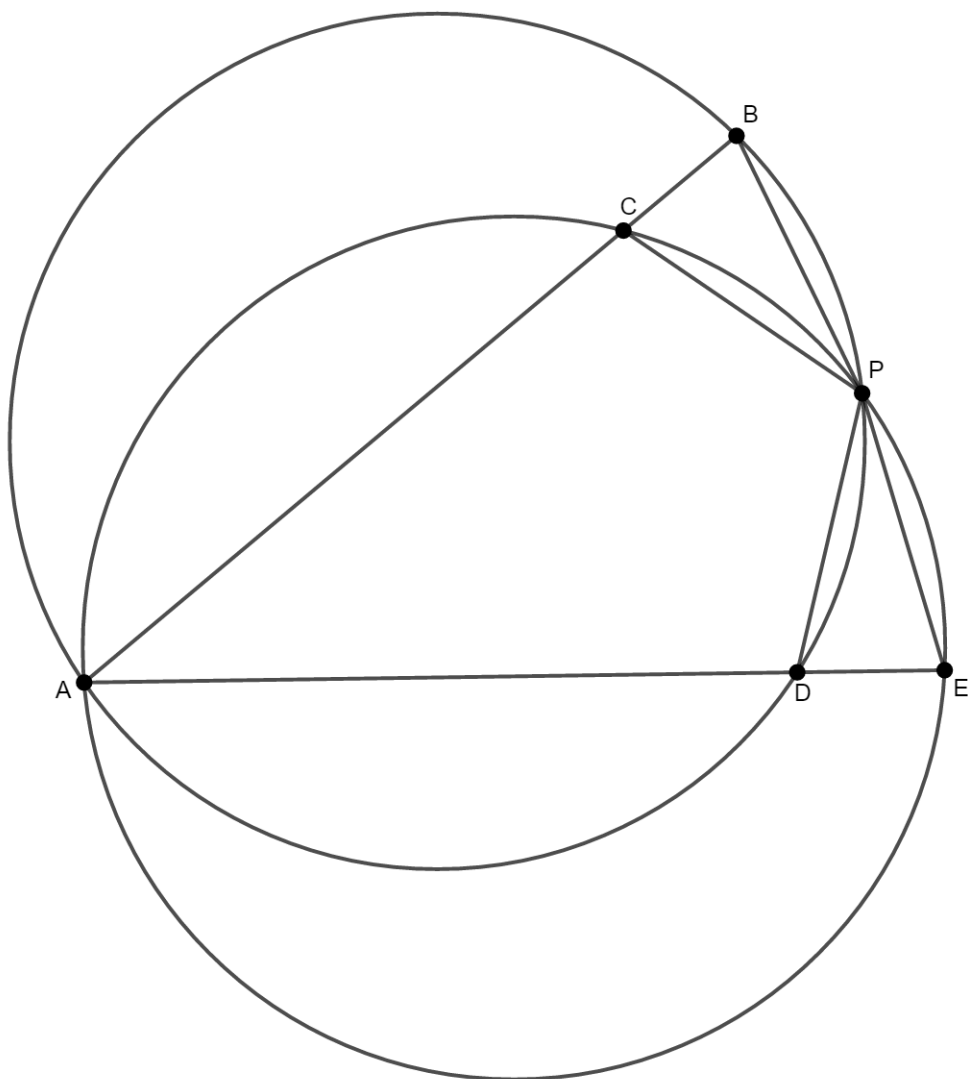


4.3.20



We have $BC = DE$, $\angle PBC = \angle PDE$, $\angle PCB = \angle PED$, so $\triangle PBC \sim \triangle PDE$, $PB = PD$ implies AP bisects $\angle BAD$.